We can continue this process for a sequence of the δ-neighbourhoods of point a $\boxed{\delta_1 = 1, \delta_2 = \frac{1}{2}, \delta_3 = \frac{1}{3}, \ldots\ldots, \delta_n = \frac{1}{n}, \ldots\ldots}$

Note that the δ-neighbourhoods are selected in such a way that the sequence (δ_n) converges to zero (is infinitesimal).

If each time we select from each δ-neighbourhood one point x in which $f(x)$ assumes a value outside of the interval $]b-\varepsilon', b+\varepsilon'[$, we obtain a sequence composed of points $\boxed{x_1, x_2, \ldots\ldots, x_n, \ldots\ldots}$

Since the sequence (δ_n) converges to zero, the sequence (x_n) inevitably converges to a. A sequence composed of the corresponding values of the function (the sequence $[f(x_n)]$) is not convergent to b because for all n we have $|f(x_n) - b| \geqslant \varepsilon'$. It means that we obtained a sequence (x_n) convergent to a for which the sequence $[f(x_n)]$ is divergent.

This contradicts the condition of the theorem which states that b is the limit of the function at a in terms of definition 2. It means that for any sequence (x_n) convergent to a the corresponding sequence $[f(x_n)]$ must be convergent to b. And the sequence (x_n) that we have found contradicts this condition.

Hence, the assumption that b, being the limit of the function in terms of definition 2, is not at the same time, the limit of the function in terms of definition 1, is invalidated. This completes the proof of the theorem.

Reader: I must admit of being wrong when I spoke about different natures of the limit of numerical sequence and the limit of function at a point.

Author: These limits differ but $\boxed{\text{their nature is the same}}$. The concept of the $\boxed{\text{limit of function at a point}}$ is based, as we have seen, on the concept of the $\boxed{\text{limit of numerical sequence}}$.

That is why basic theorems about the limits of functions are analogous to those about the limits of sequences. → V. Imp